ISOGENIES OF ELLIPTIC CURVES

In this lecture, we continue our study of isogenies of elliptic curves.

Throughout, K is a perfect field with fixed algebraic closure $\bar{K}$. Let (E_1, O_1) and (E_2, O_2) be elliptic curves defined over K. Unless otherwise stated, points are taken in $E_i(\bar{K})$.

1. Endomorphisms and automorphisms

For every integer n, the multiplication map $[n] : E \longrightarrow E$, sending $P \longmapsto [n]P$, is an endomorphism. Thus there is a natural ring homomorphism

$$\mathbb{Z} \longrightarrow \mathrm{End}(E), \qquad n \longmapsto [n].$$

It follows from an examination of the addition formulas that this is not the zero homomorphism. It follows from the same that $\mathrm{End}(E)$ is a torsion-free $\mathbb{Z}$-module.

Example 1.1. The negation map

$$[-1] : E \longrightarrow E, \qquad P \longmapsto -P,$$

is an automorphism of the elliptic curve. On a general Weierstrass equation it is given by

$$(x, y) \longmapsto (x, -y - a_1 x - a_3).$$

Example 1.2. Assume that $\mathrm{char}(K) \neq 2$, that $i \in K$ satisfies $i^2 = -1$, and let

$$E : y^2 = x^3 - x.$$

Then

$$\iota : E \longrightarrow E, \qquad \iota(x, y) = (-x, iy),$$

is an automorphism satisfying

$$\iota^2 = [-1].$$

Thus E has endomorphisms which do not arise merely from multiplication by integers. Such phenomena are the beginning of the theory of complex multiplication.

Example 1.3. Let E be defined over the finite field $\mathbb{F}_q$. The Frobenius morphism

$$\pi : E \longrightarrow E, \qquad (x, y) \longmapsto (x^q, y^q),$$

is an endomorphism. It is a purely inseparable isogeny of degree q. Although it is bijective on geometric points, it is not an isomorphism of algebraic curves: its inverse on points is not given by rational functions.

1.1. The form of an isogeny in Weierstrass coordinates.

The group-homomorphism property puts strong restrictions on the rational functions defining an isogeny.

Proposition 1.4. *Assume that* $\mathrm{char}(K) \neq 2$*, and write*

$$E_1 : y^2 = f_1(x), \qquad E_2 : y^2 = f_2(x),$$

where f_1 *and* f_2 *are cubic polynomials without repeated roots. Let*

$$\phi : E_1 \longrightarrow E_2$$

be a nonzero isogeny defined over K*. Then on the affine parts of the curves,* ϕ *has the form*

$$\phi(x, y) = \big(R(x), S(x)y\big)$$

for suitable rational functions $R, S \in K(x)$*.*

Proof. Write the affine coordinate functions of ϕ as $r_1, r_2 \in K(E_1)$. Since ϕ is a group homomorphism,

$$\phi(-P) = -\phi(P).$$

On the displayed Weierstrass equations, negation is $(x, y) \mapsto (x, -y)$. Hence

$$r_1(x, -y) = r_1(x, y), \qquad r_2(x, -y) = -r_2(x, y).$$

Every element of $K(E_1)$ can be written uniquely in the form

$$A(x) + B(x)y, \qquad A(x), B(x) \in K(x).$$

The first identity forces $r_1 \in K(x)$, while the second forces $r_2/y \in K(x)$. This gives the desired form. □

2. Kernels, degrees, and separability

Definition 2.1. Let $\phi : E_1 \to E_2$ be an isogeny. Its *kernel* is

$$\ker \phi = \{P \in E_1(\bar{K}) : \phi(P) = O_2\}.$$

Since ϕ is a group homomorphism, $\ker \phi$ is a subgroup of $E_1(\bar{K})$.

Proposition 2.2. *Let $\phi : E_1 \longrightarrow E_2$ be a nonzero isogeny. Then:*

(i) $\ker \phi$ is finite, and every fibre of ϕ is a coset of $\ker \phi$;
(ii) We have $\#\ker \phi = \deg_s \phi$;
(iii) the ramification index is independent of the point, and $e_\phi(P) = \deg_i \phi$ for every $P \in E_1$;
(iv) ϕ is separable if and only if it is unramified. In this case,$\deg \phi = \#\ker \phi$.

Proof. The kernel is the fibre above O_2, and hence is finite because ϕ is a nonconstant morphism of curves. If $\phi(P) = Q$, then $\phi^{-1}(Q) = P + \ker \phi$. Indeed, $\phi(R) = Q$ if and only if $\phi(R - P) = O_2$. Thus every fibre contains exactly $\#\ker \phi$ geometric points. On the other hand, the generic number of geometric points in a fibre is $\deg_s \phi$. It follows that

$$\#\ker \phi = \deg_s \phi.$$

For every $P \in E_1$, the homomorphism property gives a commutative diagram

$$\begin{array}{ccc} E_1 & \xrightarrow{\tau_P} & E_1 \\ {\scriptstyle\phi}\downarrow & & \downarrow{\scriptstyle\phi} \\ E_2 & \xrightarrow{\tau_{\phi(P)}} & E_2. \end{array}$$

Translations are automorphisms of curves and therefore have ramification index 1. Multiplicativity of ramification indices under composition gives

$$e_\phi(P) = e_\phi(O_1).$$

Thus the ramification index is constant. Applying the formula

$$\sum_{P \in \phi^{-1}(O_2)} e_\phi(P) = \deg \phi$$

gives $\deg \phi = \#\ker \phi \cdot e_\phi(O_1) = \deg_s \phi \cdot e_\phi(O_1)$. Hence

$$e_\phi(P) = \deg_i \phi$$

for every P. The final assertion follows immediately. □

Corollary 2.3. *If $\mathrm{char}(K) = 0$, then every nonzero isogeny is separable and*

$$\deg \phi = \#\ker \phi.$$

In positive characteristic, a nonzero isogeny may have trivial kernel without being an isomorphism. The Frobenius isogeny is the basic example.

Corollary 2.4. *Let $\phi : E_1 \to E_2$ be a separable isogeny. Then ϕ is an isomorphism if and only if*

$$\ker \phi = \{O_1\}.$$

Proof. If the kernel is trivial, then $\deg \phi = \#\ker \phi = 1$. The converse is immediate. □

Remark 2.5. If ϕ is defined over K, then its kernel is stable under the action of G_K:

$$P \in \ker \phi \quad \Longrightarrow \quad P^\sigma \in \ker \phi \qquad (\sigma \in G_K).$$

The individual points in the kernel need not be K-rational. Stability of the subgroup, rather than pointwise rationality, is the natural descent condition.

3. Constructing isogenies from finite subgroups

Every nonzero isogeny has a finite kernel. A fundamental result says that, for separable isogenies, the converse is also true.

Theorem 3.1 (Quotient by a finite subgroup)**.** *Let $E/\bar{K}$ be an elliptic curve and let $\Phi \subset E(\bar{K})$ be a finite subgroup. Then there exist an elliptic curve E' and a separable isogeny*

$$\phi : E \longrightarrow E'$$

such that $\ker \phi = \Phi$. The pair (E', ϕ) is unique up to an isomorphism of the target: if $\psi : E \longrightarrow E''$ is another separable isogeny with kernel Φ, then there is an isomorphism

$$\iota : E' \longrightarrow E''$$

such that $\psi = \iota \circ \phi$. We denote the elliptic curve E' as above by E/Φ.

If E is defined over K and Φ is stable under G_K, then E/Φ and the quotient isogeny may be chosen to be defined over K.

Sketch of proof. Each $Q \in \Phi$ acts on E by the translation automorphism

$$\tau_Q(P) = P + Q.$$

Let

$$L = \bar{K}(E)^\Phi$$

be the subfield of rational functions fixed by all these translations. The extension

$$\bar{K}(E)/L$$

is a finite Galois extension with Galois group Φ, and has degree $\#\Phi$.

By the correspondence between function fields of transcendence degree one and smooth projective curves, there is a smooth projective curve C and a separable morphism

$$\phi : E \longrightarrow C$$

whose pullback identifies $\bar{K}(C)$ with L. The fibres of ϕ are precisely the Φ-orbits. If we put

$$O_C = \phi(O),$$

then

$$\phi^{-1}(O_C) = \Phi.$$

The morphism ϕ is unramified: equivalently, the translation action of Φ on E has no fixed points except for the identity translation. Riemann–Hurwitz therefore gives

$$2g(E) - 2 = (\#\Phi)(2g(C) - 2).$$

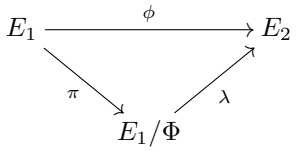
Since $g(E) = 1$, it follows that $g(C) = 1$. Thus (C, O_C) is an elliptic curve, which we call E/Φ, and ϕ is a separable isogeny with kernel Φ.

If $\psi : E \to E''$ is another separable isogeny with the same kernel, then every function in $\psi^*\bar{K}(E'')$ is invariant under translation by Φ. Hence

$$\psi^*\bar{K}(E'') \subseteq \bar{K}(E)^\Phi.$$

Both subfields have index $\#\Phi$ in $\bar{K}(E)$, so they are equal. The function-field correspondence now gives the required isomorphism of targets. The assertion over K follows by Galois descent from the stability of Φ. $\square$

Corollary 3.2. *Let $\phi : E_1 \to E_2$ be a separable isogeny, and let $\Phi \subseteq \ker \phi$ be a subgroup. Then ϕ factors through the quotient by Φ:*



for a unique isogeny $\lambda : E_1/\Phi \to E_2$.

3.1. **An explicit 2-isogeny.** The quotient construction is abstract. The following example gives an explicit formula.

Example 3.3. Assume that $\mathrm{char}(K) \neq 2$. Let $a, b \in K$ satisfy $b(a^2 - 4b) \neq 0$, and consider the elliptic curves

$$E : y^2 = x^3 + ax^2 + bx$$

and

$$E' : Y^2 = X^3 - 2aX^2 + (a^2 - 4b)X.$$

The point $T = (0, 0)$ is a point of order 2 on E. Define

$$\phi : E \dashrightarrow E'$$

on the affine part by

$$\phi(x, y) = \left(\frac{y^2}{x^2}, \frac{y(b - x^2)}{x^2} \right).$$

Using the equation of E, the first coordinate may also be written as

$$X = x + a + \frac{b}{x}.$$

A direct substitution shows that the displayed coordinates satisfy the equation of E'. Since E is smooth and E' is projective, the rational map extends to a morphism. It satisfies

$$\phi(O) = O, \qquad \phi(T) = O,$$

and no other point maps to O. Hence $\ker \phi = \{O, T\}$. Therefore ϕ is a separable isogeny of degree 2, and

$$E' \simeq E/\langle T \rangle.$$